AIDAN BRADSHAW

aidanbradshaw@cox.net | abradshaw1.github.io | abradshaw1 | aidanbradshaw1

EDUCATION

Northwestern University, Evanston, IL — Sep 2025–Present
Ph.D. in Computer Science
M.S. in Computer Science | GPA: 3.94/4.00 (Aug 2026)
Research area: Wearable time series learning for health and human behavior recognition
Advisor: Karan Ahuja

Carnegie Mellon University, Pittsburgh, PA — May 2025
M.S. in Applied Data Science | GPA: 4.1/4.0
Research area: Interpretable medical diffusion in radiology
Advisor: Adam Perer

San Diego State University, San Diego, CA — May 2024
B.S. in Computer Science (with distinction) | Minor in Mathematics | GPA: 3.81/4.0
Research area: Mobile health, human-centered computing, and multimodal sensing systems
Advisor: Shangping Ren

PROFESSIONAL EXPERIENCE

Northwestern University, Evanston, IL — Sep 2025–Present
Graduate Student Researcher | SPICE Laboratory
Transfer learning research for wearable activity recognition from IMU and vision data

Swiss Federal Institute of Technology Zurich (ETH Zurich), Zurich, Switzerland — May 2025–Jan 2026
Research Assistant | Integrated Systems Laboratory, Digital Circuits and Systems Group
Monocular pose estimation and mobile acceleration research for human movement analysis in athletes

Parsons Corporation, Pittsburgh, PA — Jan–May 2025
Artificial Intelligence Consultant
Policy based reinforcement learning with human feedback and synthetic data generation for internal LLM prompt detection

Carnegie Mellon University, Pittsburgh, PA — Aug 2024–May 2025
Research Assistant | School of Computer Science, Human-Computer Interaction Institute
Explainable AI research for text-to-image diffusion in radiology

Massachusetts Institute of Technology, Cambridge, MA — May–Nov 2024
Research Assistant | MIT Media Lab, Responsive Environments Group
Tiny machine learning and model distillation research for bio-acoustic monitoring of endangered bumblebees

AirHop Communications AI, San Diego, CA — Jun–Aug 2023
Software Engineer
Self-organizing graph network optimization for cellular communications

San Diego State Research Foundation, San Diego, CA — Mar 2023–Aug 2024
Human-Centered Computing Researcher | Computer Architecture and Systems Laboratory
Mobile health sensing and machine learning research for Raynaud's syndrome monitoring from multimodal symptom, wearable, thermal, and environmental data

AWARDS & HONORS

Murray Fellowship ($48K) | Northwestern University — 2025
Winner, Statistical Machine Learning Final Contest | Carnegie Mellon University — 2025
Best Applied Linear Models Project | Carnegie Mellon University — 2024
Elected to Phi Beta Kappa Honors Society | San Diego State University — 2024

PUBLICATIONS

Peer-Reviewed Conference Papers

6. Yiquan Li*, **Aidan Bradshaw***, Jesse Gao, and Karan Ahuja. *Towards General-Purpose Inertial Foundation Models*. **ICLR 2027** (in preparation) (*Equal contribution)
5. **Aidan Bradshaw**, Riku Arakawa, Xin Liu, and Karan Ahuja. *TransfHAR: Self-Supervised Wrist Representations for On-Demand Activity Recognition*. **ACM UIST 2026**
4. Sawyer T. Jones, **Aidan Bradshaw**, Ramaz Tskhadadze, Ben Shen, and Shangping Ren. *A Customizable, Real-Time Mobile Health Application for Raynaud's Syndrome and Beyond*. **IEEE HealthCom 2025**
3. Katelyn Morrison, Arpit Mathur, **Aidan Bradshaw**, Tom Wartmann, Steven Lundi, Afrooz Zandifar, Weichang Dai, Kayhan Batmanghelich, Motahhare Eslami, and Adam Perer. *A Human-Centered Approach to Identifying Promises, Risks, and Challenges of Text-to-Image Generative AI in Radiology*. **ACM AIES 2025**
2. **Aidan Bradshaw**, Katelyn Morrison, Arpit Mathur, Weichang Dai, Motahhare Eslami, Kayhan Batmanghelich, and Adam Perer. *Toward Interpretable 3D Diffusion in Radiology: Token-Wise Attribution for Text-to-CT Synthesis*. **MIDL 2025** (Short Paper)
1. **Aidan Bradshaw**, Ramaz Tskhadadze, Shangping Ren, and Ben Shen. *A Tailored Health Application: Monitoring the Etiology of Raynaud's Disease*. **CSCSU 2024**

Peer-Reviewed Journal Papers

2. **Aidan Bradshaw**, Elif Basokur, Marco Giordano, Luca Benini, and Christoph Leitner. *MuyBridge: Quantized 2.5D Network Fusion for On-Device Gait Estimation*. **IEEE Sensors Journal** (submitted)
1. Patrick Chwalek, Marie Kuronaga, Marco Giordano, **Aidan Bradshaw**, Isamar Zhu, Marina Arbetman, and Joseph A. Paradiso. *On-device deep learning for real-time acoustic monitoring of endangered Bombus dahlbomii and invasive congeners*. **Scientific Reports 2026**

Peer-Reviewed Poster/Demo Papers

1. Katelyn Morrison, Arpit Mathur, **Aidan Bradshaw**, Tom Wartmann, Steven Lundi, Afrooz Zandifar, Weichang Dai, Kayhan Batmanghelich, Motahhare Eslami, and Adam Perer. *Opportunities and Challenges in Designing Text-to-Image Generative AI for Medical Education, Training, and Practice*. **Pitt AI in Healthcare Research Symposium 2024**

TEACHING EXPERIENCE

Graduate Teaching Assistant, *Reasoning with Data II*, Carnegie Mellon University 2025
Graduate Teaching Assistant, *Reasoning with Data I*, Carnegie Mellon University 2024

INVITED TALKS

Guest Lecturer, Machine Learning & Sensing Course, Northwestern University 2026
Invited Panelist, Conformable Decoders, Northwestern University 2026
Guest Presenter, Feinberg School of Medicine, Northwestern University 2025

STUDENTS MENTORED

Bhavi Barnwal, Masters, *Computer Science*, Northwestern University 2026
Wenpeng Cai, Undergraduate, *Computer Science*, Northwestern University 2026
Elara Liu, Masters, *Computer Science*, Northwestern University → PhD Student @ Emory 2025

SERVICE

External Reviewer, ACM UIST, ACM IMWUT, IEEE TVCG 2026
Team Captain, SDSU Division I Club Lacrosse 2023, 2024
Facilitator and Group Lead, Orange County OCD Support Group, OCD Southern California, Gateway Therapeutic Group, USC OCD Center
Volunteer Work, Someone Cares Soup Kitchen, Aztec Rock Hunger, Breast Cancer Awareness Carnival